

F42 — substrate-Schur closure absorbing Elie
 Toy 4006: $\kappa=81/8$ FORCED (F40 closed),
 Direction-B resolved (F39 closed,
 FRAMEWORK), ε held open honestly (F38 —
 Elie’s rounding-artifact catch); P promoted to
 strong FRAMEWORK candidate, NOT
 STANDING

Lyra (Claude Opus 4.8)

2026-06-06 Saturday 14:00 EDT

F42 Substrate-Schur Closure (absorbing Elie Toy 4006)

0. Goal

Close the three substrate-Schur endpoints (F38 ε , F39 Direction-B, F40 κ) using Elie’s explicit FK color-tensored Szegő computation (Toy 4006). Honest accounting: two close, one is held open — and held open *correctly*, because Elie caught that its target is a rounding artifact.

1. F40 — $\kappa = 81/8$: CLOSED (FORCED)

Elie Toy 4006 G2–G3 forces the muon edge-term:

- G2 (rigorous): the Shilov boundary $\partial S D_{IV}^5 = S^4 \times S^1/\mathbb{Z}_2$ is a *product*, so its reproducing (Szegő) kernel factorizes, $S_{\text{Shilov}} = S\{S^4\} \cdot S_{\{S^1/\mathbb{Z}_2\}}$, and any boundary matrix element factorizes: $\kappa = \kappa_{\{S^4\}} \cdot \kappa_{\{S^1\}}$. This is my F40 structural pin, now rigorous (product $\boxtimes$ product kernel).
- G3 (geometry rigorous + color FRAMEWORK rule): $\kappa_{\{S^4\}} = N_c^{\{\dim S^4\}} = N_c^4$ (color trace once per boundary dimension; exponent 4 = $\dim S^4$ = codim-4, Casey #14 — pinned), $\kappa_{\{S^1\}} = 2^{\{-N_c\}} (\mathbb{Z}_2 \text{ quotient, } 1/2 \text{ per color})$. Product = $N_c^4 \cdot 2^{\{-N_c\}} = 81/8$.

So the muon Hardy-(1–P) boundary matrix element = $N_c^4/2^{\{N_c\}} = 81/8$ is substrate-FORCED (F32 Gate A1 / K229d closed). The geometry (factorization,

$\dim S^4 = 4$) is rigorous; the color-trace-per-dimension + $\mathbb{Z}_2$ -per-color action is FRAMEWORK-tier. Honest tier: κ CLOSED at FRAMEWORK-FORCING (the exponents pinned by geometry; the color rule is the one FRAMEWORK assumption).

2. F39 — CKM Direction-B: CLOSED (FRAMEWORK); the fork resolves

Elie G5 supplies the explicit color factors: each boundary endpoint carries one N_c color trace (per S^4), and the two endpoints give N_c^2 . This resolves the F39 fork:

$$\sin \theta_C = \frac{N_c^2}{2^{N_c} \cdot n_C} = \frac{9}{40} = \lambda, \quad N_c^2 = \underbrace{N_c}_{\text{gen-1 endpoint}} \times \underbrace{N_c}_{\text{gen-2 endpoint}}.$$

The N_c^2 in the amplitude is two independent endpoint color traces — Direction-B, not Direction-A. The two quark endpoints sit at distinct boundary points (gen-1 and gen-2 K-types), each contributing one color trace, summed independently in the boundary matrix element; the denominator $2^{\{N_c\}} \cdot n_C = 40$ is the boundary normalization. So $9/40 = \lambda$ is substrate-FORCED in form.

Honest tier and flag: this is FRAMEWORK, not RIGOROUS. It is substrate-specific — it *departs* from the SM picture, where the charged-current amplitude conserves color and carries a single trace. The claim is that the substrate boundary matrix element carries two independent traces because the endpoints are distinct boundary points not tied by a single color δ . That is supported by Elie’s boundary factorization but is a substrate structural assumption, not an SM-derived fact. F39 fork resolved toward B at FRAMEWORK tier. (And per F36: $9/40 = \lambda$ exactly, so the identification was never in doubt — only the mechanism, now Direction-B.)

3. F38 — ε : HELD OPEN (correctly), with two reasons not to fish it

Factor = $1 + \rho$; the Hardy isometry forces $\rho = 1$ (F38), so Λ factor = 2 is DERIVED — and now doubly backed: Elie’s G2 establishes the same Szegő factorization rigorously, so the bulk/boundary double-count that gives the 2 rests on rigorous geometry. This is the robust, closed content: the cosmological factor-2 is the Hardy vacuum double-count, a mechanism, not a fit.

ε = the correction. I am not closing it, for two reasons — one mine, one Elie’s:

- (mine, a self-correction): F38 asserted $\varepsilon > 0$ (“factor > 2”) from a hand-wave that the Bergman exponent (7/2) exceeds the Szegő (5/2) so the bulk is “heavier.” On reflection that sign argument is not cleanly defensible without the explicit integral — which space carries the vacuum, and in which measure, determines the sign, and the Shilov boundary (where the Hardy norm concentrates) can make $E^{\text{bdy}} \not\approx E^{\text{bulk}}$ for the opposite reason. I walk back

the F38 ε -sign claim. ε 's sign and magnitude require the explicit κ_{Bergman} -weighted per-region integral (Elie's G4 inputs: exponent gap = rank/2 = 1; $\kappa_{\text{Bergman}} = -n_C = -5$; c_FK; Shilov volume $8\pi^3/3$).

- (Elie's catch, decisive): the observed " $\varepsilon = 2.0208 - 2 = 0.0208$ " is a rounding artifact of the meV inputs ($\Lambda^{1/4}$: 4.85 / 2.4, each ~2–3 sig figs). So ε is not even *measured* to better than tens of percent — $1/48 = 0.02083$ and $1/50 = 0.02$ both sit in the noise. Fishing ε is wrong twice over: no mechanism (Cal #35) and no precise target (input rounding).

Honest tier: factor = 2 DERIVED; ε is a small correction below current theory *and* data resolution — the single genuine residual. It closes only via the explicit integral *and* better input precision. Not today, and not by fishing.

4. Cal #254 confirmed at the constant level (the real prize)

Elie 4006 confirms what F37 proposed: the muon edge-term (81/8) and the Λ factor-2 use the same Szegő factorization — literally the same projection-P / Shilov geometry computed once. The shared object across the mass sector and the cosmological sector is an operator (P), not an integer. This is “share an operator, not an integer” (Cal #254) carried all the way to the explicit constant. That, not any single number, is the substrate-Schur result.

5. Net: P promoted to strong FRAMEWORK candidate — NOT STANDING

Endpoint	Status	Tier
F40 $\kappa = 81/8$ (muon edge)	CLOSED	FRAMEWORK-FORCING (geometry rigorous; color rule FRAMEWORK)
F39 Direction-B (CKM N_c^2)	CLOSED	FRAMEWORK (substrate-specific two-trace; departs from SM)
F38 Λ factor = 2	DERIVED	Hardy isometry + Elie G2 rigorous
F38 ε correction	OPEN	needs explicit integral + better inputs; NOT fished

The Hardy/Bergman projection P is a strong, coherent FRAMEWORK candidate as a substrate-Schur generator: two sectors (muon mass, cosmological Λ) provably share its Szegő factorization. It is not promoted to STANDING — that would require (a) the color-trace-per-dimension rule derived (not assumed), (b) Direction-B's two independent traces justified beyond the substrate assumption, and (c) ε

closed. Three honest gates remain. Calling this STANDING today would be exactly the over-promotion the day's discipline has been firing on.

6. Closure

Absorbing Elie Toy 4006: F40 ($\kappa = 81/8$) closes as substrate-FORCED; F39 (CKM color²) resolves to Direction-B at FRAMEWORK tier (two independent endpoint color traces; $9/40 = \lambda$); F38's Λ factor-2 is DERIVED (Hardy isometry, rigorously backed), and ε is held open correctly — walked back the F38 sign hand-wave, and Elie's catch that the 0.0208 target is a meV rounding artifact removes any basis for fishing it. The substrate-Schur generator P is a strong FRAMEWORK candidate — muon edge and Λ factor share one Szegő factorization (Cal #254 at the constant level) — not yet STANDING, with three named gates. Two of three endpoints closed; the third honestly held. Sub-PCAP closed.

Tier: F42 sub-PCAP closure; κ FRAMEWORK-FORCING + Direction-B FRAMEWORK CLOSED; factor-2 DERIVED; ε OPEN (not fished); P = strong FRAMEWORK candidate, 3 gates to STANDING.

— Lyra, Sat 2026-06-06 14:00 EDT. F42 substrate-Schur closure absorbing Elie 4006: F40 $\kappa = N_c^4 \cdot 2^{\{-N_c\}} = 81/8$ FORCED (Shilov product factorization G2 rigorous + color-tensoring G3; exponent 4 = dim S^4 = codim-4 Casey #14 pinned); F39 Direction-B CLOSED at FRAMEWORK (N_c^2 = two independent endpoint color traces, substrate-specific departs from SM single-trace, $9/40 = \lambda$); F38 Λ factor = 2 DERIVED (Hardy isometry, Elie G2 rigorous backing) + ε HELD OPEN — walked back my F38 $\varepsilon > 0$ sign hand-wave (not defensible without explicit integral) + Elie caught the 0.0208 target is a meV rounding artifact (1/48, 1/50 in the noise), so ε not fished twice over; Cal #254 confirmed at constant level (muon-edge 81/8 + Λ -factor 2 use SAME Szegő factorization = operator P not integer); P promoted to strong FRAMEWORK candidate NOT STANDING (3 gates: color rule derivation, Direction-B justification, ε closure).